

Arduino Magnetic Door Detection System with OLED SPI Display

Alireza Naderinasab

July 4, 2026

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1 Project Description

This project implements a smart magnetic door monitoring system using an Arduino Uno, a magnetic reed switch sensor, and a 0.96-inch SPI OLED display.

The system detects whether a door is open or closed based on the presence of a magnetic field. When the magnet is near the sensor, the system detects a **CLOSED** state. When the magnet moves away, it detects an **OPEN** state.

To ensure stable readings, a software debounce technique is used to filter noise from the sensor. The detected status is displayed in real time on the OLED screen.

2 Code

The complete Arduino source code is shown below.

```
1  #include <SPI.h>
2  #include <Wire.h>
3  #include <Adafruit_GFX.h>
4  #include <Adafruit_SSD1306.h>
5
6  #define SCREEN_WIDTH 128
7  #define SCREEN_HEIGHT 64
8
9  #define OLED_CS      10
10 #define OLED_DC       9
11 #define OLED_RST      8
12
13 Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT,
14 &SPI, OLED_DC, OLED_RST, OLED_CS);
15
16 const int magneticPin = 2;
17
18 bool lastStableState = HIGH;
19 bool lastReadState = HIGH;
20
21 unsigned long lastChangeTime = 0;
22 const unsigned long debounceDelay = 50;
23
24 void drawScreen(bool closed) {
25
26     display.clearDisplay();
27
28     display.setTextSize(2);
29     display.setTextColor(SSD1306_WHITE);
30
31     display.setCursor(35, 0);
32     display.println("DOOR");
33
34     display.setCursor(10, 30);
35
36     if (closed) {
37         display.println("CLOSED");
```

```

38     } else {
39         display.println("OPEN");
40     }
41
42     display.display();
43 }
44
45 void setup() {
46
47     Serial.begin(9600);
48     pinMode(magneticPin, INPUT_PULLUP);
49
50     if (!display.begin(SSD1306_SWITCHCAPVCC)) {
51         while (1);
52     }
53
54     drawScreen(false);
55 }
56
57 void loop() {
58
59     bool reading = digitalRead(magneticPin);
60
61     if (reading != lastReadState) {
62         lastChangeTime = millis();
63         lastReadState = reading;
64     }
65
66     if ((millis() - lastChangeTime) > debounceDelay) {
67
68         if (reading != lastStableState) {
69             lastStableState = reading;
70
71             if (lastStableState == LOW) {
72                 drawScreen(true);
73                 Serial.println("Door CLOSED");
74             } else {
75                 drawScreen(false);
76                 Serial.println("Door OPEN");
77             }
78         }
79     }
80 }

```

3 Hardware Setup

The hardware used in this project consists of the following components:

- Arduino Uno

- 0.96-inch OLED SPI Display (SSD1306)
- Magnetic Reed Switch Sensor
- Jumper Wires
- Breadboard

4 Output Images

Image 1

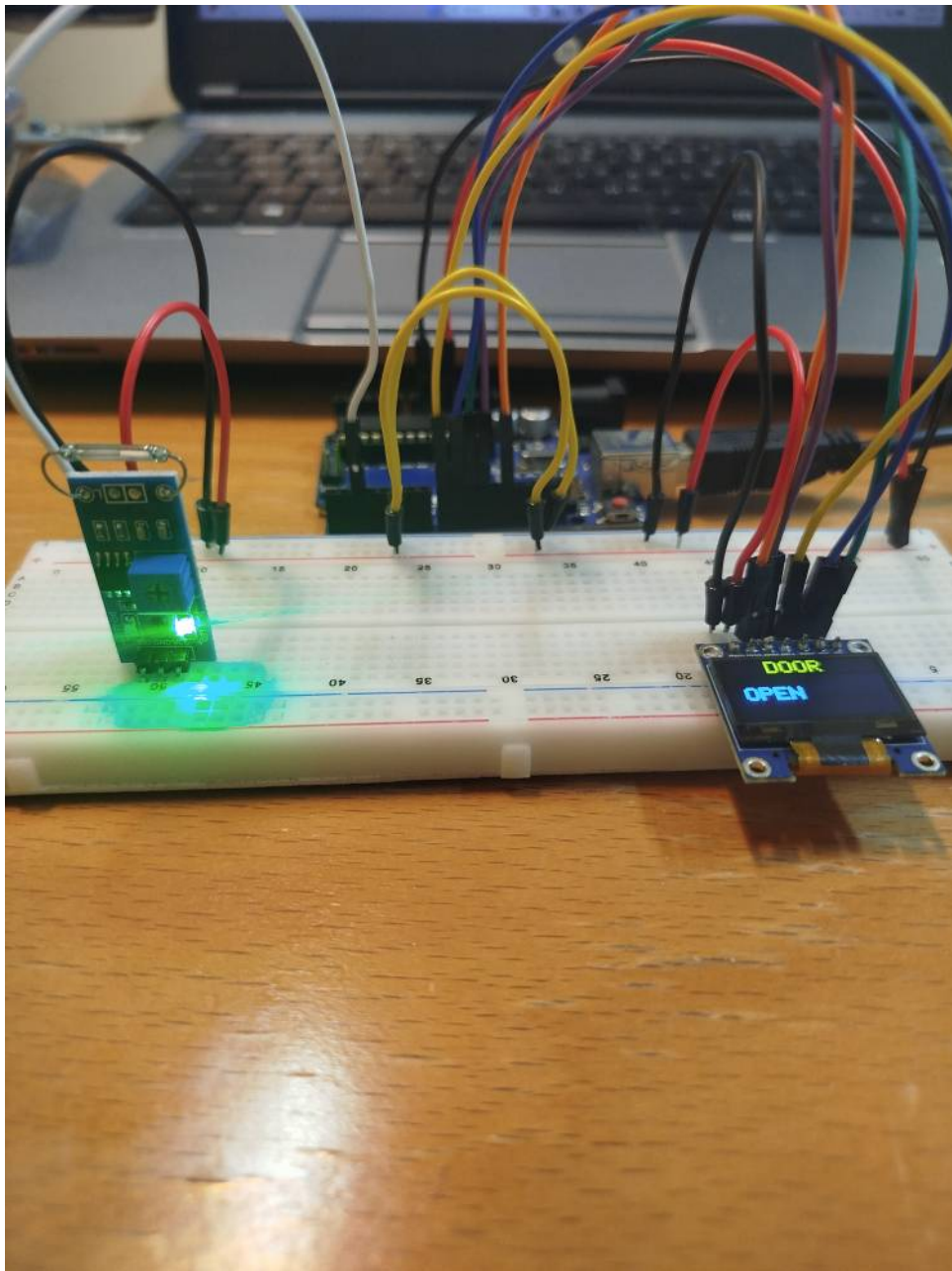
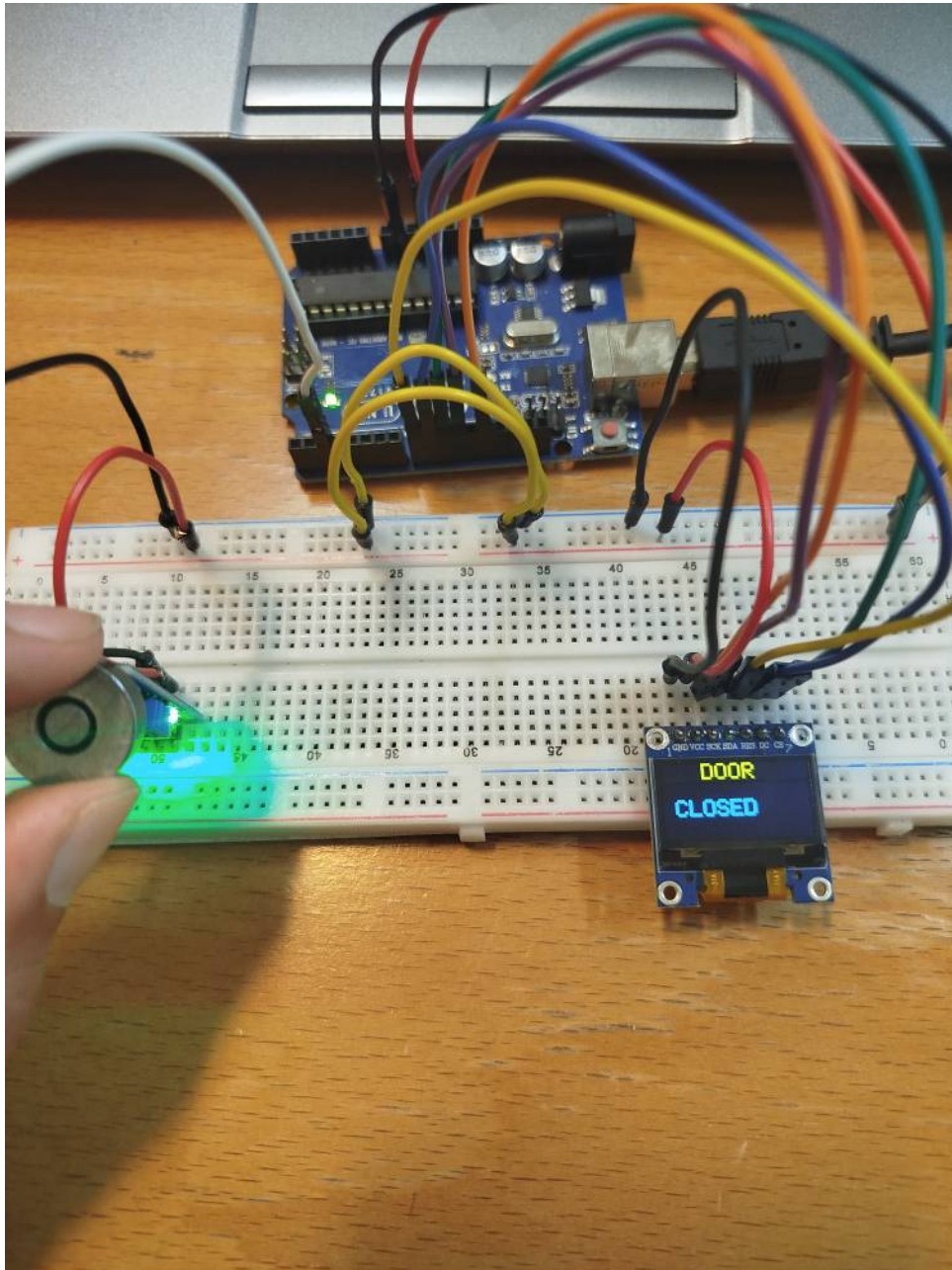


Image 2



Wiring Connections

- Magnetic Sensor → Arduino Pin 2 (with INPUT_PULLUP)
- OLED VCC → 5V
- OLED GND → GND
- OLED SCK → Pin 13
- OLED MOSI → Pin 11
- OLED CS → Pin 10

- OLED DC $\rightarrow$ Pin 9
- OLED RST $\rightarrow$ Pin 8

5 System Behavior

- Magnet near sensor $\rightarrow$ CLOSED state
- Magnet away from sensor $\rightarrow$ OPEN state
- Debounce filtering ensures stable output
- OLED updates only when state changes (no flicker design)

6 Conclusion

This project demonstrates a simple but robust embedded system for door monitoring using magnetic sensing and SPI OLED display.

It introduces:

- Sensor interfacing
- SPI OLED communication
- Debounce algorithm for noise reduction
- Real-time embedded display systems

This system can be extended for smart home security, IoT monitoring, and alarm systems.